

Pranjal Arya

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Skills

- **Languages:** C, ARM assembly.
- **Protocols / Device drivers:** PCIe (SR-IOV), USB, SD card, SDIO, UART, SPI, and I2C.
- **Processor:** ARM Cortex v7 and v8 based SoCs.
- **Tools:** Git, GDB, Saleae analyzer, SEGGER SystemView.
- **Miscellaneous:** Linux, FreeRTOS, TrustZone for Cortex-M.

Open source contribution

📄 MbedTLS: cryptographic hardware acceleration library

Experience

Qualcomm, Hyderabad

Member of a core team responsible for the development of a complete PCIe software stack for Qualcomm custom SoCs.

- **Senior Software Engineer** May 2024 - Present
 - Developed PCIe Gen 5.0 endpoint driver from scratch to enable custom Qualcomm multi-core SoC based on Arm v8 cores.
 - Enabled SR-IOV to enable multiple virtual machines on the host.
 - Enabled MHI transport layer protocol over PCIe 5x16 link to communicate with Linux host running KMD drivers.
 - Contributed to features that resulted in 8% performance and 14% power enhancement of the overall link.

Infineon Technologies, Bangalore

Member of the team responsible for the development of Device drivers, middleware & firmware for Infineon PSoC MCUs.

- **Senior Software Engineer** April 2023 - April, 2024
 - Participated in code review and helped new team members ramp up.
 - Device drivers, firmware & bring-up:
 - * Performed pre- and post-silicon bring-ups on various ARM Cortex-M4 and M33 based SoC.
 - * Developed firmware and BSP for upcoming PSoC6 SoC using different toolchains.
 - * Ensured smooth integration of delivered assets to middlewares, RTOS.
- **Software Engineer** July 2020 - March 2023
 - Device drivers:
 - * Designed and developed features for USB 2.0, SD Card, SDIO, UART, SPI, I2C, and GPIO device drivers.
 - * Analyzed critical issues reported by customers and internal teams and provided fixes.
 - Middleware:
 - * Involved in USB - HID middleware development.
 - * Upgraded cryptographic hardware acceleration to enable MbedTLS release in public GitHub (<https://bit.ly/prarya>).
 - BSP and CI: Created scripts and makefiles to automate build and basic sanity testing. Integrated this framework in CI using Git.
- **Intern** January 2020 - June 2020
 - Internship project: Unified device drivers to generate a common library to enable a family of PSoC6 devices from a single source.

Education

B.Tech. (Electronics & Communication Engineering) - Nirma University, Ahmedabad

CPI: **8.24** | 2016 - 2020

HSC

88.9% | 2016

SSC

89.7% | 2014

Projects

- **Custom Real-Time Operating System development from scratch** 📄 GitHub project link

Developed a custom RTOS kernel and added support for Cortex-M architecture with the following features:

 - Round-Robin scheduler.
 - Heap memory manager supporting aligned memory allocation and reporting maximum memory usage statistics.
 - Spinlock and semaphore support for synchronization.
 - Mailbox for communication.

- **Custom bootloader**

 [GitHub project link](#)

Developed a custom bootloader for STM32L476RG MCU with the following features:

- Host application in PC interacts with a custom bootloader over UART.
- Can maintain version information and update application firmware using an A/B swap mechanism.
- Integrated a secure boot mechanism to verify the authenticity of the firmware image.

- **CPU Faults on Arm Cortex-M**

 [GitHub project link](#)

To understand ARM processor internals, generated various Arm Cortex-M4 system exceptions using software on TM4C123GXL Tiva C series MCU. Wrote a fault handler to dump the processor status over debug prints.

Awards

- **Global Spot Award**

Infineon Technologies, January 2023

Awarded for significant contribution to device driver library development.

- **YIP Award (Infineon Idea Management Program)**

Infineon Technologies, February 2023

Awarded to a team of 2 members for internal framework improvement resulting in a 40% cost reduction.